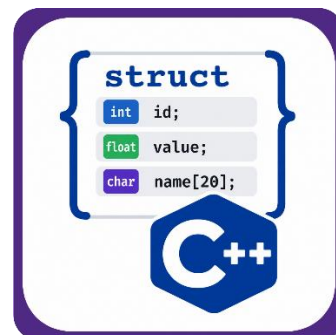
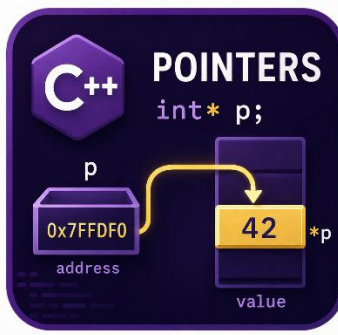
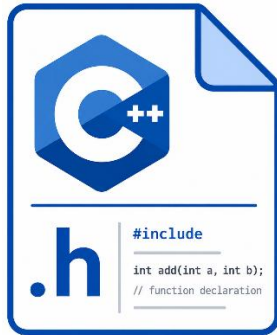




LAB 1

FALL 2026

CONCEPTS



DESCRIPTION

- This lab is basically a review of structures, pointers, and header files that you learned near the end of CSC 1300.
- You will be building a structure named **Course** that will hold information about a course at Tennessee Tech.
- Then you will implement functions to create a course, destroy a course, and print a course.
- Then, you will complete a driver that will use the Course structure and functions.

POLICIES FOR LAB 1

AI POLICY FOR LAB 1: NO GENERATIVE AI

Generative AI including GitHub Copilot can't be used to generate or suggest code in this lab. If you generate part or all of your code with GenAI, you will be charged with an Academic Integrity Violation.

WORKING WITH OTHERS

- This is not a paired programming lab.
- You may not send your code to any other student (unless it is a lab instructor during tutoring).
- You are encouraged, however, to help each other debug/troubleshoot and show a few lines of code to other students in your lab.

LATE ASSIGNMENT POLICY

- Lecture and lab programming assignments may be completed/submitted up to 2 days late at a penalty of 10 points off per day late.
- After 2 days late, the assignment may not be submitted, and the grade will be a zero. Emailed submissions are not accepted.

POLICY CONCERNING WHAT YOU CAN'T USE IN PROGRAMMING ASSIGNMENTS

You may not use code or syntax in your programming & lab assignments that have not been taught in the class without referencing your source. **You may not ever use the following in your programs:**

- `while(true)` or `while(1)`
Check out <https://sites.tnitech.edu/acrockett/2025/08/15/why-is-whiletrue-or-while1-bad-programming-practice/> to see why this is bad programming practice.
- `goto` statements
Check out <https://sites.tnitech.edu/acrockett/2025/08/15/why-is-goto-bad-programming-practice-in-c/> to see why this is bad programming practice.
- recursively calling `main`

- `exit()`
- ternary operators (any syntax requiring a '?')
- range-based for loops
- try-catch
- `getline(cin>>ws, variable);`
- lambda functions
- defining all programmer-defined functions above the main function
- code that has no comment block at the top indicating the filename, author, date, and purpose

If you use any of the things listed above or use code that has not been taught in the class (without referencing your detailed source in a comment) then full points will be deducted for that criteria item in the rubric.

GRADING POLICY

There is a **rubric** attached to the lab assignment in Canvas that details how you will be graded. Make sure to review the rubric prior to the due date to understand our expectations and to ensure you do not miss any points unnecessarily.

SPECIFICATIONS

Build a structure to contain **Course** information and the functions described below.

COURSE STRUCTURE

The structure tag should be **Course** and should be defined in **Course.h**

- `Name (string)`
- `Location (string)`
- `Sections (pointer that will eventually point to an array of strings)`
- `Number of sections (integer)`
- `Number of credit hours (integer)`

FUNCTIONS

Function prototypes should go in **Course.h** and function definitions should go in **Course.cpp**

CREATECOURSE FUNCTION

Purpose:

This is a function that will (dynamically) create a new Course structure variable and then enter the given data into the Course structure members. This function will also have to dynamically create the sections array containing the number of elements as there are sections of the course. Then the function will return the memory address of the Course structure variable from this function.

Function Prototype:

```
Course* createCourse(string, string, int, int );
```

Parameters:

- a string containing the name of the course
- a string containing the location of the course
- an integer containing how many sections the course has
- an integer containing how many credit hours the course has

Returns:

- a pointer to the Course variable just created with all the given data

DESTROYCOURSE FUNCTION

Purpose:

This is a function that will delete both the dynamically created sections array and the course so that there are no memory leaks.

Function Prototype:

```
void destroyCourse(Course* myCourse)
```

PRINTCOURSE FUNCTION

Purpose:

This function will print all the data in the members of the sent Course structure variable, including all the sections of the course. The data must be printed in a neat easy-to-read format so that the program is user friendly.

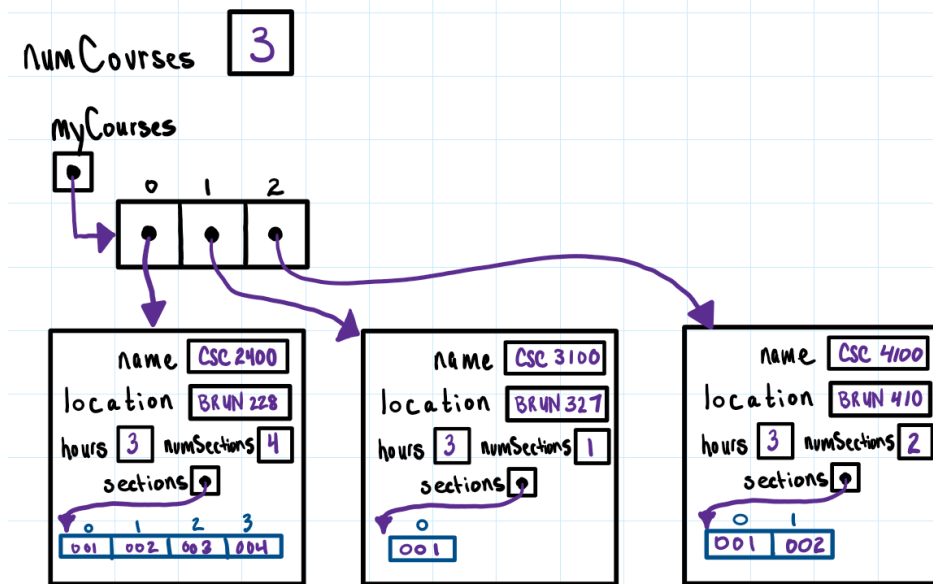
Function Prototype:

```
void printCourse(Course* myCourse)
```

DRIVER

You are given most of the **Driver.cpp** code (called Driver_given_lab1.cpp and you will need to rename it), but **you will need to add only a few LINES** of code in all the places indicated by the comments in the code.

Below is a drawing of an example of how the data is organized in the program.



After you add the necessary code, you need to test all the code to make sure you get the same output as in the sample output in this document.

TEST YOUR PROGRAM!

- Does it compile?
- Have you tried entering in exactly what is in sample output (at bottom of this assignment)? This is how you will be graded.
- Do you use consistent indentions?
- Do you have some comments throughout your code? Do you have a comment above each function with the function name and purpose?
- Do you have a comment block in all of your files with the filename, author, date, and purpose of program?
- Is your code readable and neat or does it have weird whitespace all throughout it?

FILL OUT A LAB REPORT

You will fill out this lab report for every lab and it is part of your grade. To get credit, you must upload a screenshot of the confirmation page to this lab assignment.

Name your screenshot **lab1ReportProof**. (any file type as long as instructor can open & see image)

Lab Report Link: https://tntech.co1.qualtrics.com/jfe/form/SV_0iJcCTCBgguPDpQ

HOW TO TURN IN

Zip all the following files and upload to Canvas in the Lab 1 assignment under Module 1.

- Driver.cpp
- Course.h
- Course.cpp
- Lab1ReportProof

SAMPLE OUTPUT

All user input is highlighted in yellow.

How many courses are you taking this semester?

3

COURSE NAME: CSC 2400

COURSE LOCATION: BRUN 228

COURSE HOURS: 3

NUMBER OF SECTIONS? 4

SECTION 1: 001

SECTION 2: 002

SECTION 3: 003

SECTION 4: 004

COURSE NAME: CSC 3100

COURSE LOCATION: BRUN 327

COURSE HOURS: 3

NUMBER OF SECTIONS? 1

SECTION 1: 001

COURSE NAME: CSC 4100

COURSE LOCATION: BRUN 410

COURSE HOURS: 3

NUMBER OF SECTIONS? 2

SECTION 1: 001

SECTION 2: 002

The following are the courses you entered:

***** COURSE 1*****

COURSE NAME: CSC 2400

COURSE LOCATION: BRUN 228

COURSE HOURS: 3

COURSE SECTIONS:

001

002

003

004

***** COURSE 2*****

COURSE NAME: CSC 3100

COURSE LOCATION: BRUN 327

COURSE HOURS: 3

COURSE SECTIONS:

001

***** COURSE 3*****

COURSE NAME: CSC 4100

COURSE LOCATION: BRUN 410

COURSE HOURS: 3

COURSE SECTIONS:

001

002